

Day 11 - Preprocessing Pipeline

Cleaning Applied

Based on the previous data quality analysis, no missing values were found in the dataset, so no missing value treatment was required.

The duplicate rows were kept because they represent valid repeated ADC readings.

The bit column was removed from the input features because it was found to be identical to the labels column. Keeping this column could give the model direct information about the correct answer.

Train, Validation, and Test Split

The dataset was divided into training, validation, and test datasets using a stratified split.

The final split was:

- Training data: 5,958 rows (70%)
- Validation data: 1,277 rows (15%)
- Test data: 1,277 rows (15%)

The stratified split helped maintain the class distribution in all three datasets.

The class balance remained approximately:

- Label 0: 62.5%
- Label 1: 37.5%

This shows that the data was divided properly without significantly changing the original class distribution.

Feature Scaling and Encoding

The feature column contains received signal values that are already normalized between 0 and 1.

Therefore, no additional feature scaling was applied at this stage.

The labels column already contains binary values of 0 and 1, so no additional label encoding was required.

Saved Processed Data

The processed datasets were saved separately as:

- processed_data/train.csv
- processed_data/val.csv
- processed_data/test.csv

Saving these files will allow me to use the same data split in the upcoming Week 3 modelling experiments.

This will also make the experiments easier to reproduce without creating a new data split every time.